

10 Convolutional networks

$$10.1 \quad f[x_i] = z_i = w_1 x_{i-1} + w_2 x_i + w_3 x_{i+1}$$

$$\text{w. } f[f[x_i]] = f[f[x_i]] \text{ with } f[i] = i-s$$

$$\begin{aligned} f[f[x_i]] &= w_1 x_{i-s-1} + w_2 x_{i-s} + w_3 x_{i-s+1} \\ &= w_1 f[x_{i-1}] + w_2 f[x_i] + w_3 f[x_{i+1}] \\ &= f[f[x_i]] \quad \square \end{aligned}$$

$$10.2 \quad z_1 = w_1 \cdot 0 + w_2 x_1 + w_3 x_2$$

$$z_2 = w_1 x_2 + w_2 x_3 + w_3 x_4$$

$$z_3 = w_1 x_4 + w_2 x_5 + w_3 x_6$$

$$z_4 = w_1 x_6 + w_2 x_7 + w_3 x_8$$

$$\Rightarrow z_i = w_1 x_{2(i-1)} + w_2 x_{2(i-1)+1} + w_3 x_{2(i-1)+2}$$

$$10.3 \quad z_i = w_1 x_{i-2} + w_2 x_i + w_3 x_{i+2}$$

$$10.4 \quad z_i = \sum_{m=-r}^r w_{m+r+1} x_{si+dm} \quad \text{with } r = \frac{k-1}{2}, \text{ kernel size } k, \text{ stride } s, \text{ dilation } d$$

$$z_i = w_1 x_{3i-9} + w_2 x_{3i-6} + w_3 x_{3i-3}$$

$$w_4 x_{3i} + w_5 x_{3i+3} + w_6 x_{3i+6} + w_7 x_{3i+9}$$

10.5 x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8

h_1 x 0

h_2 $\backslash$ x 0

h_3 $\backslash$ x 0

h_4 $\backslash$ x 0

Q5 10.3a
b

x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8

h_1 x 0 $.$

h_2 $\backslash$ x 0 $.$

h_3 $-$ $\backslash$ x 0 $.$

h_4 $-$ $\backslash$ x 0 $.$

h_5 $-$ $\backslash$ x 0 $.$

h_6 $-$ $\backslash$ x 0 $.$

h_7 $-$ $\backslash$ x 0

h_8 $-$ $\backslash$ x

Q5 10.3c

x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8

h_1 x 0

h_2 x 0

h_3 $\backslash$ x 0

h_4 $\backslash$ x 0

h_5 $\backslash$ x 0

h_6 $\backslash$ x 0

h_7 $\backslash$ x

h_8 $\backslash$ x

Q5 10.3d

10.6	x_1	x_2	x_3	x_4	x_5	x_6	
h_1	X	0					
h_2	\	X	0				
h_3		\	X	0			
h_4			\	X	0		
h_5				\	X	0	
h_6					\	X	Per 10.5a-b
h_7	.	-					
h_8		.	-				
h_9			.	-			
h_{10}				.	-		
h_{11}					.	-	
h_{12}						.	

10.7 - | \ + 0 .

	h_1	h_2	h_3	h_4	h_5	h_6	h_7	h_8	h_9	h_{10}	h_{11}	h_{12}
h_1'		\					0 .					
h_2'	-		\				+ 0 .					
h_3'		-		\				+ 0 .				
h_4'			-		\				+ 0 .			
h_5'				-		\				+ 0 .		
h_6'					-						+ 0	

10.8! C_{in} input channels, C_{out} output channels, k kernel size

$$|w| = C_{out} \cdot C_{in} \cdot k, |b| = C_{out}$$

as stride, padding, and dilation do not change param. count:

1st layer: $3 \cdot 4 \cdot 3 = 36$ weights, 4 biases

2nd layer: $10 \cdot 4 \cdot 5 = 200$ weights, 10 biases

10.9 $RF_1 = 3, RF_2 = 5, RF_3 = 7$

$\begin{matrix} \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{matrix} \quad \begin{matrix} \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{matrix} \quad \begin{matrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{matrix}$

10.10 $RF_1 = 7, RF_2 = 7 + (7-1) = 13$
 $RF_3 = 13 + 6 = 19$

10.11 $RF_1 = 5, RF_2 = 9, RF_3 = 25$

$\begin{matrix} \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{matrix} \quad \begin{matrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{matrix}$

10.12 dataset size N (training examples, batch size B , number of optimizer steps (iterations) S , number of epochs E

then: $E = \frac{SB}{N}$ here: $\frac{100000 \cdot 100}{4000} = 2500$

10.13

	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}	x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	x_{26}	...
h_{11}	w_{22}	w_{23}					w_{32}	w_{33}					
h_{12}	w_{21}	w_{22}	w_{23}				w_{31}	w_{32}	w_{33}				
h_{13}		w_{21}	w_{22}	w_{23}			w_{31}	w_{32}	w_{33}				
h_{14}			w_{21}	w_{22}	w_{23}			w_{31}	w_{32}	w_{33}			
h_{15}				w_{21}	w_{22}	w_{23}			w_{31}	w_{32}	w_{33}		
h_{16}					w_{21}	w_{22}				w_{31}	w_{32}		
h_{21}	w_{12}	w_{13}					w_{22}	w_{23}					
h_{22}	w_{11}	w_{12}	w_{13}				w_{21}	w_{22}	w_{23}				
h_{23}		w_{11}	w_{12}	w_{13}			w_{21}	w_{22}	w_{23}				
h_{24}			w_{11}	w_{12}	w_{13}			w_{21}	w_{22}	w_{23}			
h_{26}				w_{11}	w_{12}	w_{13}			w_{21}	w_{22}	w_{23}		
h_{26}					w_{11}	w_{12}				w_{21}	w_{22}		
...													

10.14 $25 \cdot 3 \cdot 10 = 750$

10.15 i.e., $x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \ x_8$

h_1 | / - \

h_2 | / - \

h_3 | / -

...

⊗

$x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \ x_8$

h_1 / -

h_2 | / -

h_3 | / -

h_4 | / -

weight matrix

or kernel

size three,

stride one

//

$x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \ x_8$

h_1 / -

h_2 | / -

h_3 | / -

h_4 | / -

weight matrix

or kernel size

three, stride two

$$1016 \text{ 1st conv: } 11 \cdot 11 \cdot 96 \cdot 3 \quad (\text{weights})$$

$$+ 96 \quad (\text{bias})$$

$$2\text{nd conv: } 96 \cdot 5 \cdot 5 \cdot 256$$

$$+ 256$$

$$3\text{rd conv: } 3 \cdot 3 \cdot 256 \cdot 384$$

$$+ 384$$

$$4\text{th conv: } 3 \cdot 3 \cdot 384 \cdot 384$$

$$+ 384$$

$$5\text{th conv: } 3 \cdot 3 \cdot 384 \cdot 256$$

$$+ 256$$

$$\text{fully connected layer 6: } 9216 \cdot 4096 \quad \left(n_{in} \cdot n_{out} \text{ weights, } n_{out} \text{ bias} \right)$$

$$+ 4096$$

$$-11- \quad 7: \quad 4096 \cdot 4096$$

$$+ 4096$$

$$-11- \quad 8: \quad 4096 \cdot 1000$$

$$+ 1000 \quad \sum \quad 62 \quad 378 \quad 344$$

$$10.17 \quad r_e = (r_{e-1} - 1) s_e + k_e \Leftrightarrow r_{e-1} = s_e \cdot r_e + k_e - s_e \quad r_0 = 1$$

$$\text{here: } r_1 = (1-1) \cdot 4 + 11 = 11, r_2 = (11-1) \cdot 2 + 3 = 23, r_3 = (23-1) \cdot 1 + 5 = 27$$

$$r_4 = (27-1) \cdot 2 + 3 = 55, r_5 = (55-1) \cdot 1 + 3 = 57$$

10.18 same logic as above, in total 138 357 544 params.

10.19 Each weight of a fully connected layer with m inputs is sampled from a Gaussian distribution $w_{ji} \sim \mathcal{N}(0, \frac{2}{m})$.